ERP & INDUSTRY 4.0 — FROM STRATEGY TO EXECUTION

Individual Assignment

HIGH-LEVEL ERP SOLUTION DESIGN

Bburago Group Pvt Ltd.

Auto-Component Manufacturing and Assembly Factory — Machining, Fabrication and Assembly

Proposed Platform: Oracle Fusion Cloud ERP

Student Name	Naman Jain
Roll Number	FT272022
Programme	PGPM 2026–27
Institute	Great Lakes Institute of Management
Course	ERP & Industry 4.0 — From Strategy to Execution
Industry Selected	Manufacturing
Company (Illustrative)	Bburago Group Pvt Ltd.
Proposed Platform	Oracle Fusion Cloud ERP and SCM
Total Marks	10

Academic assumptions disclaimer. *Bburago Group Pvt Ltd. is a hypothetical company created for this classroom assignment. All operating parameters, plant configurations, volumes, organisation roles, tolerances and approval thresholds are stated assumptions used to make the ERP design concrete; they are not drawn from any real company, published report or benchmark study. No external data source is cited and none is implied.*

Statement on benefits. *Every improvement referred to in this report is illustrative and directional. It describes the capability the designed configuration would make possible, not a measured, quantified or guaranteed outcome. No percentage savings, ROI figure or performance benchmark is claimed anywhere in this document.*

1. Industry Overview, Operating Context and ERP Objectives

Bburago Group Pvt Ltd. runs an auto-component manufacturing and assembly factory, machining and fabricating components and building them into sub-assemblies for vehicle OEMs and Tier-1 customers. Production mixes make-to-order against customer release schedules with make-to-stock for high-runner part numbers, and two process steps — heat treatment and plating — are sent outside. Growth has been capacity-led and each plant automated on its own, so item structures, operation times and product costs now live in spreadsheets that disagree with each other. The proposal is a single Oracle Fusion Cloud ERP and SCM instance operated as one system with one source of truth, so that Manufacturing, Supply Planning, Inventory Management, Procurement, Order Management, Financials with Cost Management, and HCM all read and write the same Fusion data model.

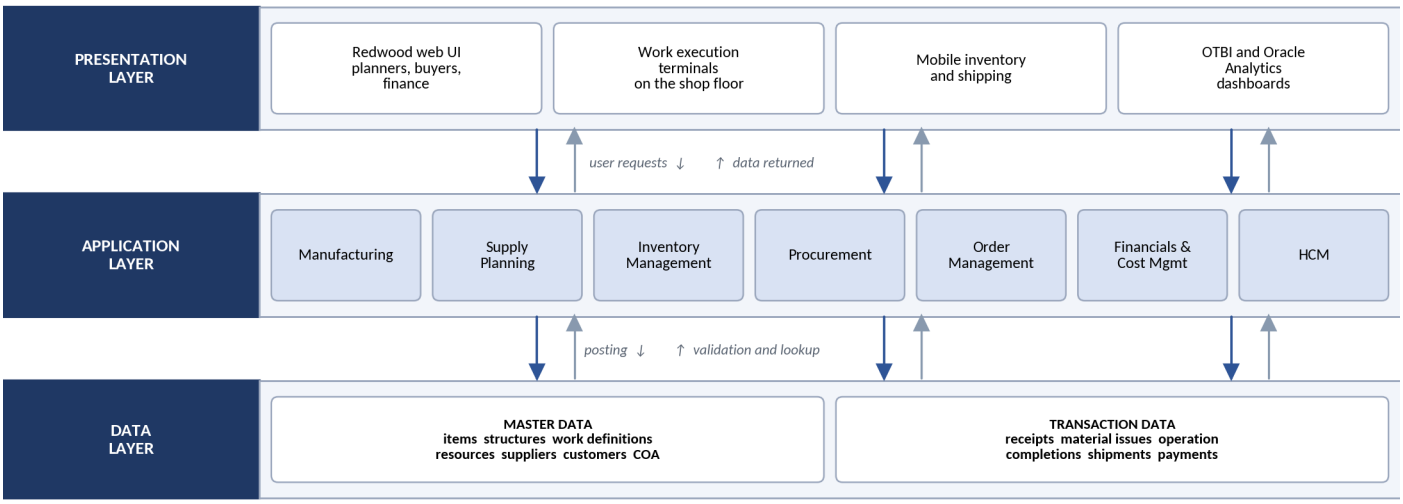
1.1 Assumed Operating Profile (classroom assumptions)

Plants	2 inventory organizations, 1 central finished-goods store	**Item records**	Approx. 9,000 across raw material, components, WIP and finished goods
Production type	Discrete — machining, fabrication, assembly; 3 shifts	**Suppliers**	Approx. 350, incl. steel, castings, fasteners and outside-processing suppliers
Employees	Approx. 1,400, majority on the shop floor	**Customers**	Approx. 40 OEM and Tier-1 accounts on release schedules
Planning mode	Make-to-order plus make-to-stock for high runners	**Statutory**	GST-registered in multiple states; Apr–Mar financial year

1.2 Current Operational Problems and the Corresponding ERP Objective

Current problem (as-is)	ERP objective (to-be)
Stores, work-in-process and finance each hold a different inventory figure; WIP is estimated at period end.	One item master and one on-hand position across every inventory organization, subinventory and locator, including material at outside processing.
Item structures and operation times are maintained in plant spreadsheets; engineering changes reach the shop floor late or not at all.	Item structures and work definitions held centrally in Oracle Product Hub, versioned and released through change orders with effective dates.
Requisitions for direct material are approved over e-mail with no budget check and no check of a qualified source.	Sequential mandatory approval workflow in Approvals Management with budgetary control, qualified-supplier validation, escalation and audit trail.
Supplier invoices are matched by hand against paper receipt notes, causing disputes and delayed payments.	Automated three-way match of purchase order, receipt and supplier invoice in Oracle Payables before payment release.
Product cost is a periodic spreadsheet exercise; material and resource variances are not visible by part number or line.	Standard cost held in a cost book, with material, resource and overhead variances posted from every work order by Cost Management.
Customer release schedules are re-keyed into a planning spreadsheet and material planning is run outside the system.	Schedules received into Order Management and consumed by Supply Planning, which nets requirements into purchase and work orders.
Functional silos: planning, plant operations and finance each work from their own version of the truth.	Integrated end-to-end processes — Order to Cash, Procure to Pay, Plan to Produce, Record to Report and Hire to Retire — on shared master data.

1.3 Three-Layer ERP Architecture



Oracle Fusion Cloud ERP is delivered as software as a service on one Fusion data model. A module is a functional view of the same data, not a separate application holding its own copy.

2. Required ERP Modules and Ranking of the Five Most Critical (Q1 — 2 Marks)

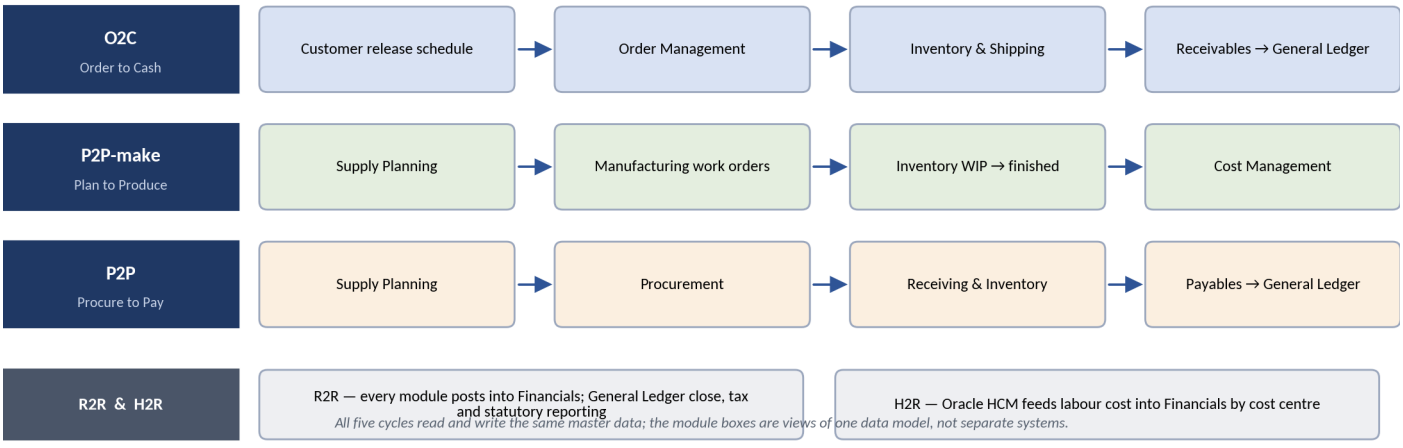
Rank	Module (Oracle Fusion Cloud)	Manufacturing-specific justification for Bburago Group
1	**Manufacturing — work definitions and work execution**	This is where the factory adds value and where cost is created. Work definitions, work orders, operation completions, scrap, rework and outside-processing operations all sit here. Without it the shop floor stays on paper job cards, work in process is an estimate, and no other module can reconstruct what was built from what was consumed. Ranked first because it is the only module that models the business the company is in.
2	**Supply Planning (Planning Central and Demand Management)**	OEM release schedules move weekly. Supply Planning explodes those schedules through the item structure, nets them against on-hand, WIP and open supply, and releases planned orders as purchase and work orders. It is the module that converts a customer commitment into executable work, and the difference between planning in a spreadsheet and planning in the system.
3	**Inventory Management (with Shipping and Receiving)**	Four stock states must be tracked distinctly: raw material in stores, work in process on the line, material at an outside-processing supplier, and finished goods awaiting shipment. Lot and serial genealogy is also a customer requirement in auto components — a field complaint must be traceable back to a material lot and a work order.
4	**Procurement (Self Service Procurement, Purchasing, Supplier Model)**	Direct material is the single largest element of product cost. Blanket agreements, qualified-source control, outside-processing purchase orders and disciplined receiving determine both gross margin and whether the line runs at all.
5	**Financials — General Ledger, Payables, Receivables and Cost Management**	Standard cost and variance analysis by part number and line, work-in-process valuation, multi-state GST compliance and statutory reporting. Financials is the only module that can hold the single financial truth, because every other module ultimately posts into it.

Why Order Management ranks just below the top five. It is essential and is in Phase 1 scope, but demand at Bburago Group arrives as a small number of customer release schedules rather than as high-volume order capture. Those schedules are consumed by Supply Planning, so the order-entry burden is comparatively low — which is why the module is core but not among the five most critical.

2.1 Complete Module Landscape

Core (Phase 1)	Manufacturing; Supply Planning; Inventory Management with Shipping and Receiving; Procurement; Order Management; Financials (General Ledger, Payables, Receivables, Assets, Tax) with Cost Management.
Supporting	Product Hub for item and structure governance with change orders; Quality Management for inspection plans and non-conformance; Maintenance for plant assets; Oracle HCM for worker records, shifts and competencies; Expenses; OTBI and Oracle Analytics for reporting; Supplier Portal for schedule and invoice collaboration.
Future scope	Machine-level IoT and OEE capture from the shop floor; production scheduling optimisation; predictive maintenance on critical machines; supplier collaboration extended to capacity signals; digital-twin simulation of the assembly line.

2.2 Mapping to End-to-End Process Cycles



3. Key Master Data Required Before Any Transaction (Q2 — 2 Marks)

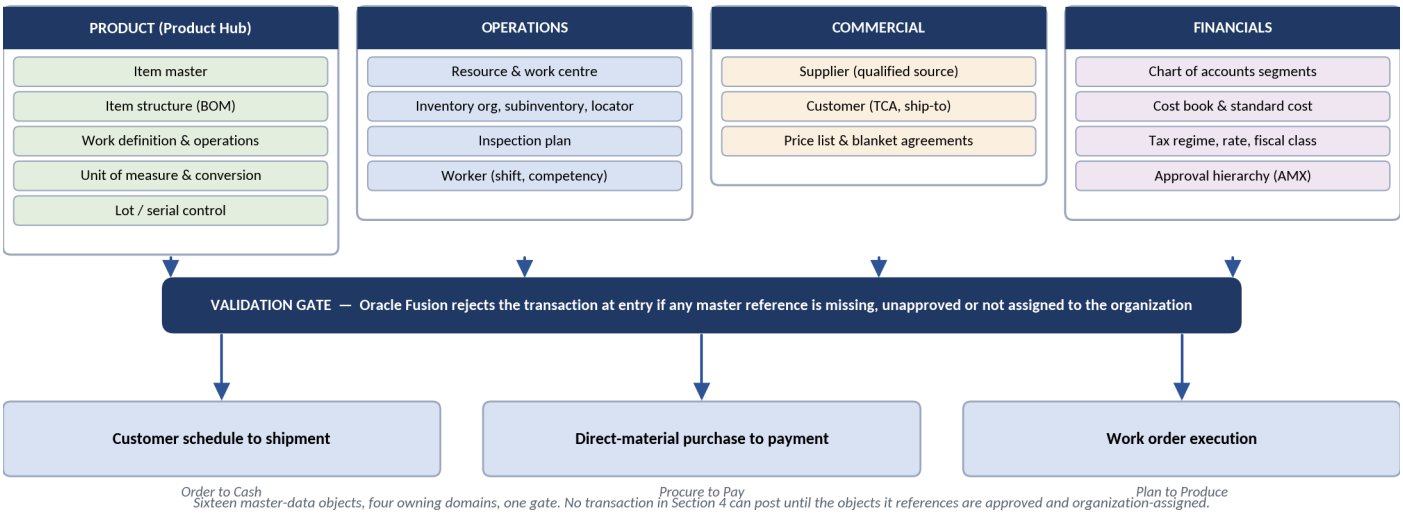
Master data	Created once, changes rarely, reused by many transactions. It describes a thing that exists — an item, an item structure, a work definition, a machine, a supplier, an account segment. It carries no date of occurrence.
Transaction data	A dated business event — a receipt, a material issue, an operation completion, a shipment, a payment. It always references master data and cannot be posted if the master record it points to is missing, unapproved or not assigned to the organization.
Why sequence matters	Oracle Fusion rejects the transaction at entry when its master reference fails validation. In a manufacturing implementation this is the hardest gate of all, because item structures, work definitions and standard costs must be right before a single work order can be released.

#	Master-data object (Oracle Fusion Cloud)	Owning function	Where it is used / why it must exist first	Consequence if the data is wrong or missing
1	**Item master (Product Hub)**	Product Engineering with Product Hub governance	Every purchase, material issue, work order, inspection, shipment and count references the item. Carries item class, item organization assignment and costing attributes.	A wrong item class or organization assignment sends the item down the wrong planning and costing logic; nothing downstream can be corrected without reversing every transaction posted against it.
2	**Item structure (bill of material) and structure version**	Product Engineering (Design)	Defines what goes into the part and in what quantity. Supply Planning explodes it; the work order draws component requirements from it.	Planning nets the wrong requirement, so either the line stops for a missing component or money is spent on material that is not needed. Product cost is wrong from the first level upward.
3	**Work definition — operations, resources and standard times**	Manufacturing Engineering	Defines the operation sequence, the work centre for each operation and setup and run time. Drives capacity planning, shop-floor execution and resource absorption.	Capacity is planned against fictional times, delivery promises slip, and resource and overhead cost is absorbed at the wrong rate so reported margin is unreliable.
4	**Resource, work centre and shift calendar**	Manufacturing Engineering with Plant Operations	Represents machines, labour and tooling capability with shift calendars and planned downtime. Supply Planning schedules against it.	The plan loads machines that are unavailable and under-loads those that are free; bottlenecks stay invisible until the schedule has already been missed.
5	**Supplier master (Supplier Model, incl. bank and GSTIN)**	Procurement, with Finance verification	Required before a purchase order can be issued, a receipt recorded or an invoice paid. Carries terms, lead time, tax registration and qualification status.	Duplicate suppliers split spend and defeat negotiated terms; a wrong bank account sends payment to the wrong party; buying from an unqualified source creates a customer-facing exposure.
6	**Customer master (Trading Community Architecture)**	Order Management with Finance credit check	Identifies who is supplied, at which ship-to site, on what commercial terms and against which blanket sales agreement or release schedule.	Shipments reach the wrong plant gate or are applied to the wrong schedule line, which for an OEM customer triggers a line-stoppage claim rather than a polite query.
7	**Inventory organization, subinventory and locator**	Plant Operations with IT	Defines where stock physically sits and in which state. Drives available to promise, material issue, outside-processing tracking and shipping.	Stock appears in stores when it is already on the line or at a supplier, so planning sees material that cannot be used and the plant buys it twice.
8	**Unit of measure and conversion**	Product Engineering with Supply Chain	Reconciles buying units (kilogram, metre, coil) with consumption and selling units (each, set) so material arithmetic holds across the chain.	A coil received is counted as an each, or the reverse; on-hand balance, consumption per unit and every downstream valuation are corrupted at source.
9	**Lot and serial control attributes**	Quality Management with Product Engineering	Decides which items are lot-controlled or serial-controlled, which is how a finished part is traced back to a material lot and a work order.	Genealogy breaks. A customer complaint cannot be narrowed to an affected lot, so containment becomes a full-population recall instead of a targeted one.
10	**Cost book and standard cost (material, resource, overhead rates)**	Finance — Cost Management	Values material issues, absorbs resource and overhead cost, and provides the baseline against which every variance is measured.	Inventory and work in process are valued at a stale standard; purchase price and usage variances are meaningless, so nobody can say whether a part is profitable.
11	**Chart of accounts segments (plant, cost centre, product line)**	Finance — Controller	Every posting from every module lands in an account combination built from these segments. This is the backbone of the General Ledger.	Transactions post to suspense or to the wrong segment, so plant-level and product-line profitability cannot be reported and the close is delayed.
12	**Tax setup — regime, rates, fiscal classifications**	Finance — Taxation	Determines the tax applied on each purchase, outside-processing movement and sales line, and drives statutory return filing.	Incorrect tax is charged or claimed, input tax credit is lost on mismatched documents, and outside-processing movements attract avoidable statutory questions.
13	**Price list, blanket purchase agreement and blanket sales agreement**	Sourcing and Order Management, with Finance sign-off	Supplies the purchase price on the order and the selling price on the customer schedule, each with validity dates.	Orders are placed at unagreed prices and invoices raised at superseded ones; margin leaks quietly and the mismatch only surfaces at three-way match.

#	Master-data object (Oracle Fusion Cloud)	Owning function	Where it is used / why it must exist first	Consequence if the data is wrong or missing
14	**Worker record — shift, competency, cost centre (Oracle HCM)**	Human Resources	Identifies who completed an operation, who approved a requisition and whose labour cost sits in which cost centre. Underpins segregation of duties.	Approval tasks route to people who have left; operation completions cannot be tied to a shift or a competency; labour cost lands in the wrong cost centre.
15	**Inspection plan and specification (Quality Management)**	Quality Assurance	Defines what is checked, on which sample size, against which tolerance, at receiving, in-process and final inspection.	Material is accepted without the checks the customer was promised, or good material is rejected on the wrong specification; either way the quality record cannot be defended in a customer audit.
16	**Approval hierarchy and authority limits (Approvals Management)**	Finance with Human Resources	Supplies the workflow in Section 5 with who may approve what, up to which value, and who acts during a recorded absence.	Requisitions route to vacant positions or stall entirely; escalation cannot resolve, and approvals get taken outside the system to keep the line running.

3.1 Sequencing — Master Data That Must Exist Before Each Transaction in Section 4

Customer schedule to shipment	Direct-material purchase to payment	Work order execution
Item master and finished-goods definition; customer and ship-to site in TCA; price list and blanket sales agreement; inventory organization and finished-goods subinventory; unit of measure; tax setup; lot or serial control; chart of accounts segments with the plant value.	Supplier with GSTIN, bank details and qualification status; item and purchase unit of measure; blanket purchase agreement price; inventory organization, subinventory and receiving parameters; inspection plan; payment terms; approval hierarchy; chart of accounts segments with the cost centre.	Item structure and structure version; work definition with operations and standard times; resource, work centre and shift calendar; unit-of-measure conversions; WIP subinventory; cost book and standard cost rates; worker record with shift and competency; in-process inspection plan.



Sixteen master-data objects, four owning domains, one gate. No transaction in Section 4 can post until the objects it references are approved and organization-assigned.

3.2 Master-Data Governance Rules Adopted

Single point of creation Each object has exactly one owning function that may create it; plants may request, never create.	Maker–checker Item, structure, work definition, standard cost, supplier and bank records require a second, different approver before release.	Change orders and effective dates Structures, work definitions and cost scenarios are versioned through change orders; nothing is edited in place once transactions exist against it.	De-duplication and inactivation Name, GSTIN and part-number duplicate checks at entry; records are made inactive rather than deleted so history survives.

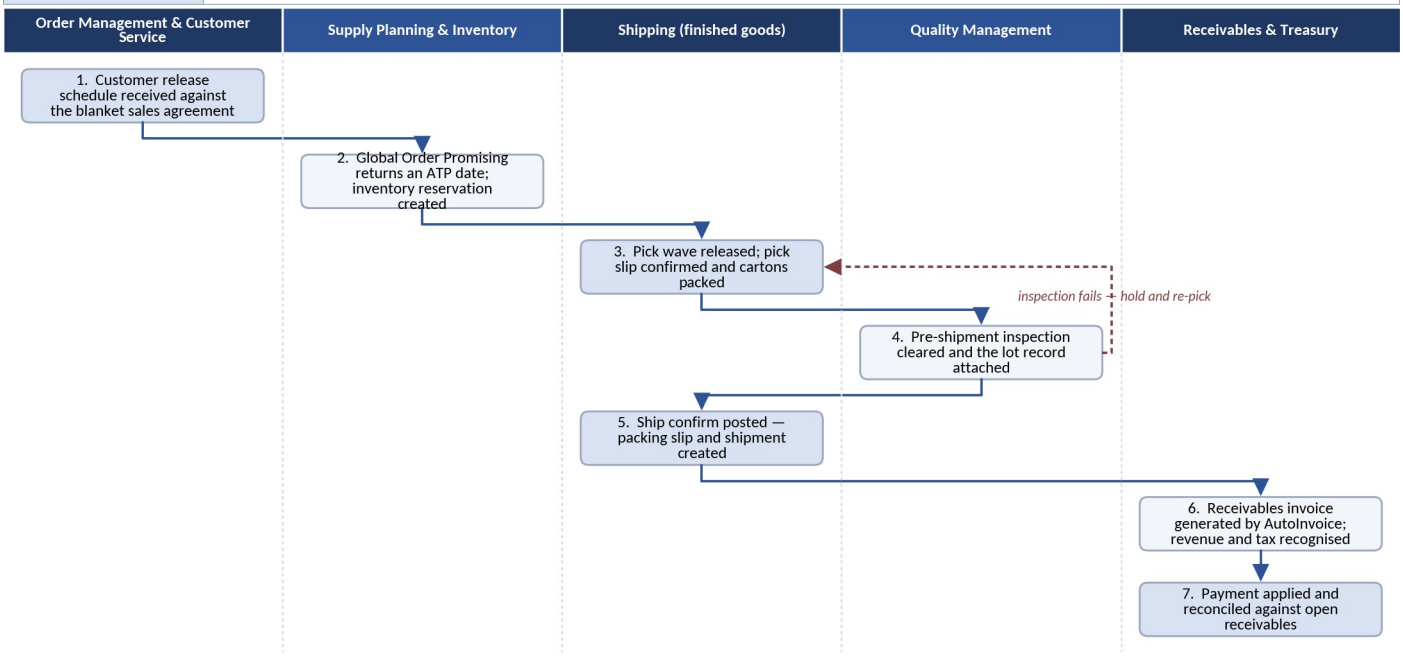
3.3 Data-Quality Dimensions Monitored, and What Each One Protects at Bburago Group

Dimension	What is measured, and the specific manufacturing exposure if it is not
Completeness	Proportion of active records with every mandatory attribute populated. An item without a cost record, or a work definition without a standard time, stops a transaction at the moment it is most inconvenient — at work order release.
Uniqueness	Duplicate records per master object. Duplicate part numbers split demand across two structures so planning nets neither correctly; duplicate suppliers fragment negotiated spend.
Accuracy	Agreement between the record and physical or contractual reality — structure quantity to what the operator actually consumes, standard time to what the machine actually takes, bank account to supplier. This is the dimension that causes direct financial loss rather than inconvenience.
Consistency	The same value means the same thing in every module: one unit-of-measure convention, one part-numbering standard, one cost-centre segment. Inconsistency is how silos re-form inside a single system.
Validity and timeliness	Records conform to the allowed format and range, and changes take effect on the correct date. A change order released late means parts continue to be built to a superseded revision — a timeliness failure with a scrap and warranty cost.

4. Three High-Frequency Business Transactions (Q3 — 2 Marks)

4.1 Transaction 1 — Customer Release Schedule to Shipment and Invoice (Order to Cash)

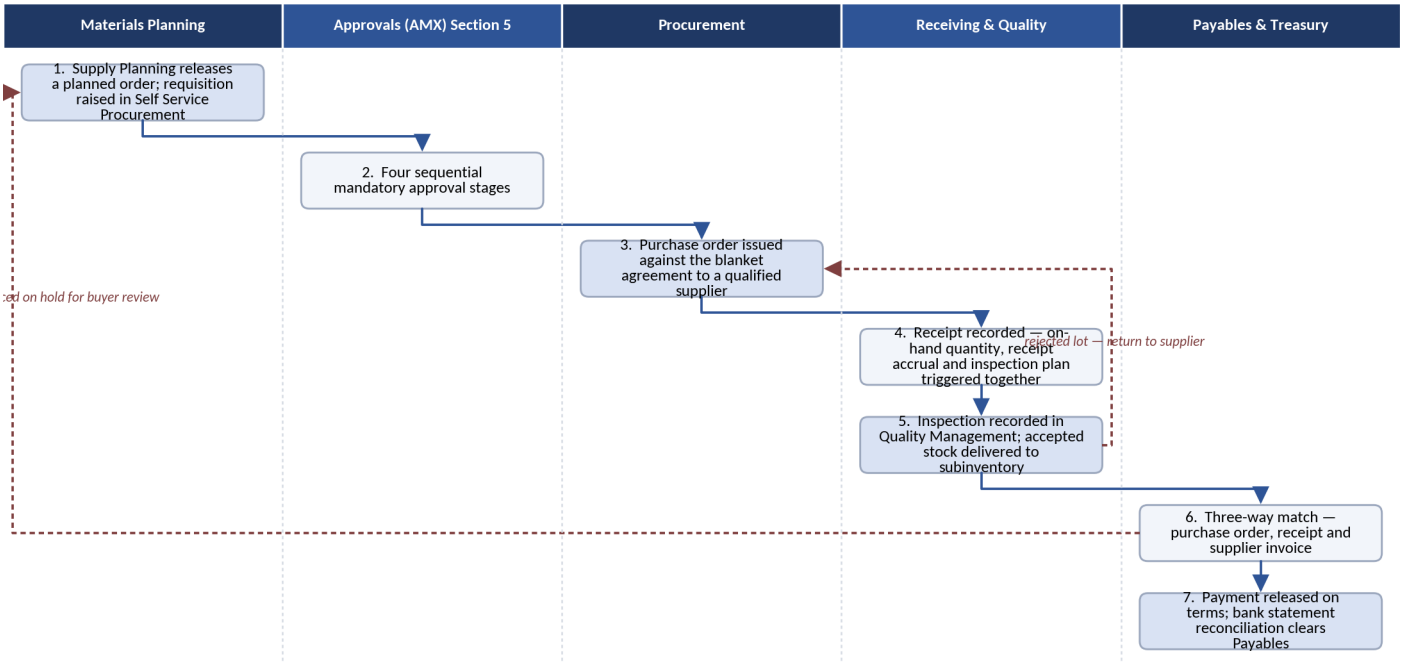
Departments	Order Management and Customer Service; Supply Planning; Shipping; Quality Management for pre-shipment clearance; Finance — Receivables and Treasury
Modules	Order Management → Inventory Management (Reservations, Shipping) → Financials (Receivables, General Ledger)
Records created	Sales order lines against the blanket sales agreement, inventory reservation, pick wave and pick slip, shipment and packing slip, Receivables invoice, statutory shipping document reference, receipt application and General Ledger journals



The reservation is relieved at ship confirm; until then the quantity is removed from available to promise without moving physically.

4.2 Transaction 2 — Direct-Material Purchase to Payment (Procure to Pay)

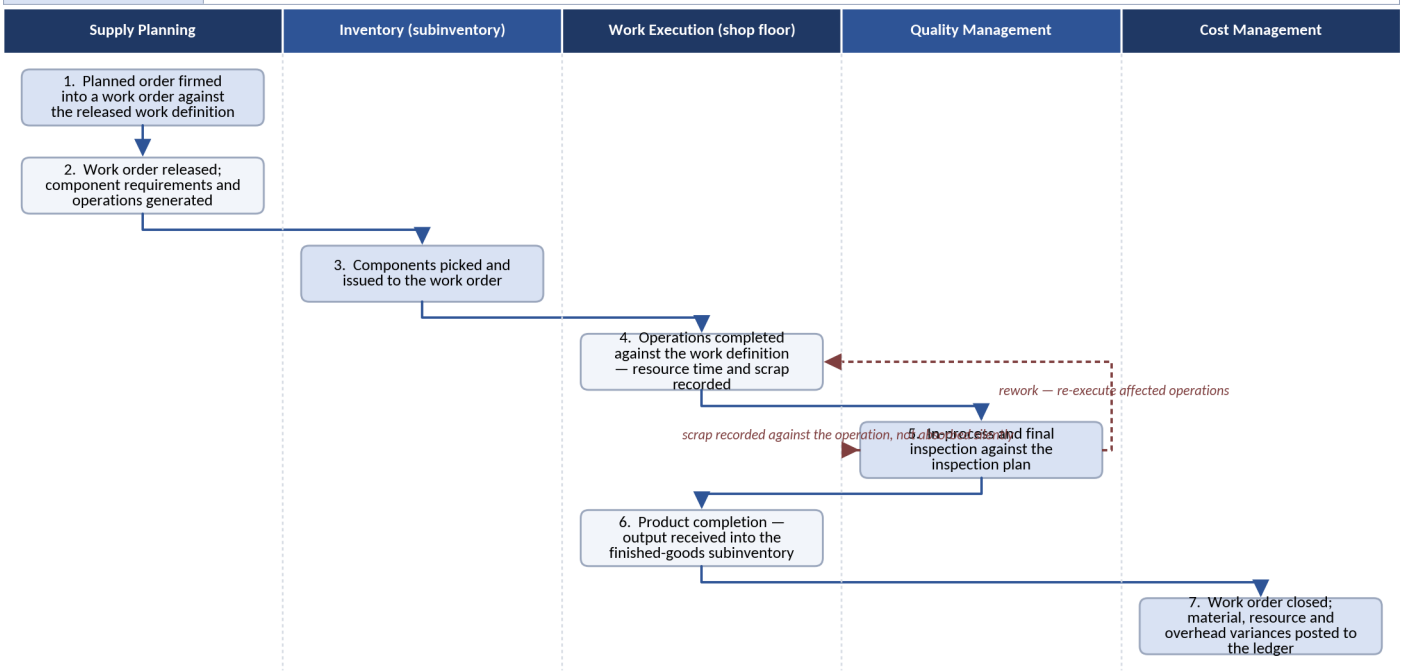
Departments	Plant Materials and Planning; Sourcing; Receiving stores; Quality Assurance; Finance — Payables
Modules	Supply Planning → Procurement → Inventory Management (Receiving) → Financials (Payables, General Ledger)
Records created	Requisition, purchase order against the blanket agreement, receipt and delivery to subinventory, inspection result, supplier invoice, three-way match outcome, receipt accrual and payment



Outside processing follows the same path: the operation is bought as an OSP item and the company's own material stays on its books while off site.

4.3 Transaction 3 — Work Order Execution (Plan to Produce)

Departments	Supply Planning; Inventory stores (material issue); Work Execution supervisors and operators; Quality Management; Finance — Cost Management
Modules	Supply Planning → Manufacturing (Work Execution) → Inventory Management → Financials (Cost Management, General Ledger)
Records created	Work order against the released work definition, component requirements and material issue transactions, operation completions with resource time and scrap, inspection results, product completion receipt, work order close with variance journals



Operation-pull and assembly-pull supply types keep the physical and the financial record of the same event in step.

4.4 Key Control Concepts Embedded in the Three Transactions Above

Concept	How it works in the Bburago Group design
Available to promise (ATP)	Global Order Promising answers on-hand less reserved, plus confirmed supply and planned production inside the horizon, so a release line is only committed to a date the factory can actually meet.
Reservation	A soft claim that removes quantity from available to promise without moving it physically. Held against a sales order until ship confirm, and against a work order until components are issued.
Receipt (Receiving)	Recorded against a specific purchase order line. It is the point at which on-hand quantity, the receipt accrual for goods received but not invoiced, and the inspection requirement all come into existence together.
Three-way match	Purchase order price and quantity, receipt quantity, and supplier invoice value must agree within tolerance. A mismatch places the invoice on hold for buyer review instead of paying it and arguing later.
Product completion and work order costing	The posting event that turns shop-floor activity into accounting reality — component consumption relieved from stores, resource and overhead absorbed at standard, work in process relieved, output received at standard cost, and the resulting variances posted to the General Ledger rather than estimated at period end.
Payment reconciliation	Two streams: outbound supplier payments matched to the bank statement to clear Payables, and inbound customer receipts applied to open Receivables invoices so the receivable position is reliable at any point in the period.

Scope note. *Point-of-sale posting is not applicable to this operating model, since Bburago Group ships to OEM customers on schedule rather than selling over a counter. The equivalent posting events in a manufacturing design are the product completion above and the ship confirm and invoice postings in Transaction 1; both are included in place of it.*

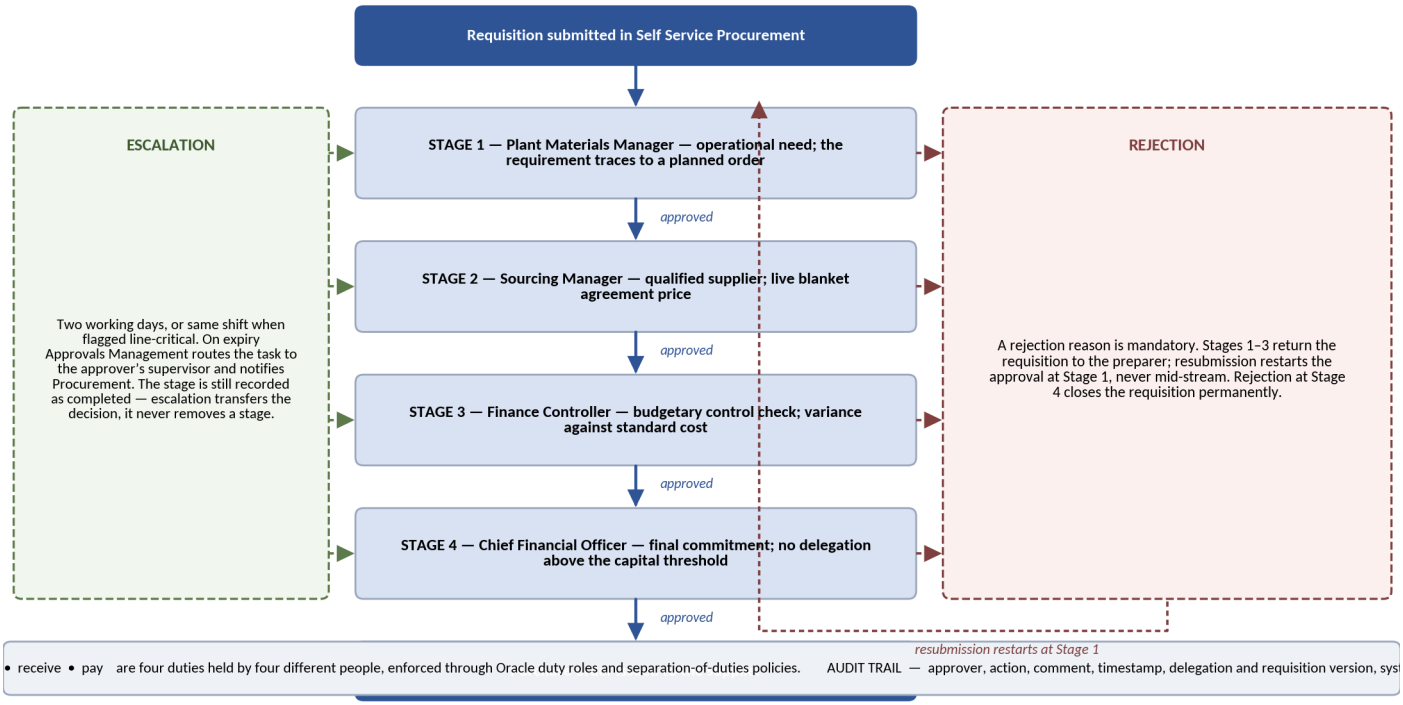
4.5 Why These Three Qualify as the High-Frequency Transactions

Transaction	Assumed frequency	Basis for selecting it
Customer schedule to shipment and invoice	Daily shipments against weekly OEM release schedules	The highest-value transaction in the business and the only one the customer sees. All revenue and every receivable originate here, and a missed shipment becomes an OEM line-stoppage issue rather than a lost sale.
Direct-material purchase to payment	Daily receipts across roughly 350 suppliers, including outside processing	Direct material is the largest element of product cost and the largest cash outflow. It also carries the greatest control exposure, which is why the workflow in Section 5 sits on it.
Work order execution	Continuous across three shifts on every machining and assembly line	The highest-count internal transaction, and the one place where cost is created rather than merely recorded. Errors here misstate work in process, product cost and margin simultaneously.

5. Critical Approval Workflow — Requisition to Purchase Order (Q4 — 2 Marks)

This workflow is chosen because it is the point at which Bburago Group commits money and commits to a source. It is configured in Oracle Approvals Management as a requisition approval flow with four sequential, mandatory stages. Each stage generates an approval task for a named approver; the requisition cannot advance until that stage is completed, and no stage can be skipped or run in parallel.

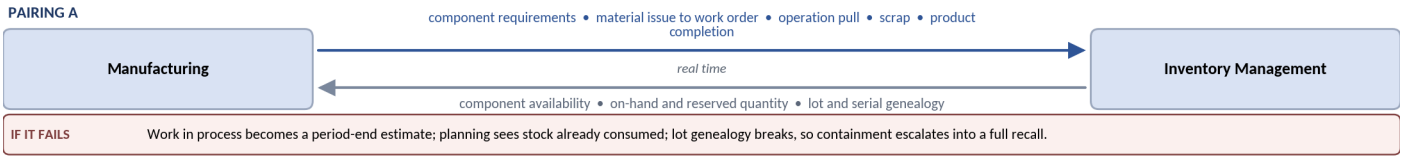
Stage	Approver role	Purpose of the stage	Approval rule applied	What the approver can and cannot do
1	**Plant Materials Manager (preparer's reporting line)**	Operational need and quantity sanity	Mandatory for every requisition, any value. Confirms the requirement traces to a planned order released by Supply Planning and that the quantity respects lot size and shelf life.	May approve, reject with a reason, or return for edit. May not prepare and approve the same requisition.
2	**Sourcing Manager**	Commercial terms and validity of the source	Mandatory. Validates that the supplier is qualified for the part with a current approval status, and that the price matches the live blanket purchase agreement.	May approve or reject. May not change the price on the requisition; a price change forces a new requisition.
3	**Finance Controller**	Budgetary and standard-cost control	Mandatory. Budgetary control validates the plant and cost-centre budget; a price beyond an assumed tolerance against standard cost is flagged for written justification.	May approve, reject, or hold pending budget reallocation. May not amend quantity.
4	**Chief Financial Officer (delegate: Chief Operating Officer)**	Final commitment authority	Mandatory for every requisition in this design, with an assumed capital threshold above which no delegation is permitted at all.	Final approval converts the requisition into a purchase order. Rejection closes the requisition permanently.

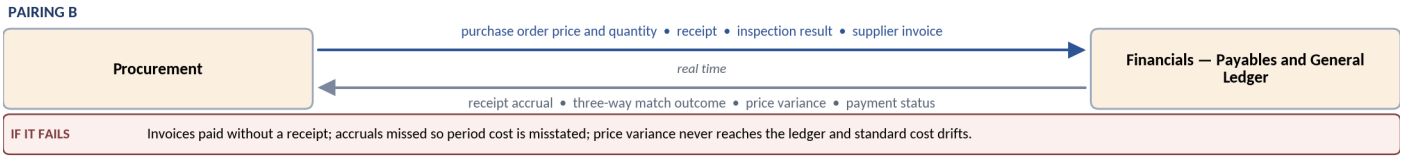


Rejection handling	A rejection reason is mandatory. Stages 1–3 return the requisition to the preparer with a reason code; the preparer may correct and resubmit, which restarts the approval at Stage 1 rather than resuming mid-stream. Rejection at Stage 4 closes the requisition and a fresh one is required.
Escalation	Each approval task carries an assumed two-working-day expiry, reduced to same-shift for a requisition flagged line-critical. On expiry Approvals Management routes the task to the approver's supervisor and notifies Sourcing. Escalation transfers the decision but never removes the stage — the stage is still recorded as completed by the escalated approver. Planned absence is handled through a recorded vacation rule, not by sharing credentials.
Segregation of duties	Four duties are held by four different people: preparing the requisition, approving it, recording the receipt in stores, and releasing the supplier payment. Oracle duty roles and separation-of-duties policies enforce this as a conflict, so a user cannot hold an incompatible pair. This is the control that stops one individual creating a supplier, buying from it, receiving nothing and paying for it.
Audit trail	The approval history stores, for every stage, the approver identity, action taken, comment, date and time stamp, any delegation or escalation applied, and the requisition version approved. The record is system-written and cannot be edited by business users; it is retained after the purchase order is issued so the commitment traces back to the person who authorised it — which is also what a customer or statutory audit asks to see.

6. Two Module Pairings Requiring Tight Integration (Q5 — 2 Marks)

	Pairing A — Manufacturing ↔ Inventory Management	Pairing B — Procurement ↔ Financials (Payables and General Ledger)
Data exchanged	Component requirements from the item structure; material issued from stores to the work order; operation-pull and assembly-pull consumption; scrap and rework quantity; product completion into finished goods; material sent to and returned from outside processing; lot and serial genealogy.	Purchase order price, quantity and tax; receipt quantity and date; inspection result; supplier terms and bank details; invoice value; receipt accrual for goods received not invoiced; purchase price variance against standard cost; payment status.
Direction	Bi-directional. Inventory confirms component availability to Manufacturing; Manufacturing posts the consumption, scrap and completion that change it.	Bi-directional. Procurement supplies the commitment and the receipt; Financials returns payment status and invoice-hold feedback.
Trigger	Work order release, each operation completion, and product completion.	Receipt recorded in Receiving, followed by entry of the supplier invoice.
Timing	Real time on material issue and product completion; pull consumption at operation completion; work-in-process relief at work order close.	Real-time accrual on receipt; three-way match at invoice entry; payment on the agreed terms date.
Purpose	One physical truth across stores, work in process, outside processing and finished goods, so what the plan assumes is what the shop floor actually holds.	Pay only for what was ordered, received and inspected, and recognise the liability and the price variance in the period they arose.
Consequence if integration fails	Work in process becomes a period-end estimate. Material issued to the line is not relieved from stores, so Supply Planning sees stock already consumed and stops replenishing until the line halts, or buys it twice. Scrap is absorbed silently into consumption, so yield cannot be measured. Most seriously for an auto-component supplier, lot and serial genealogy breaks, so a customer complaint cannot be narrowed to an affected lot and containment escalates into a full recall.	Invoices are paid without a matching receipt, or correct invoices sit unpaid and a supplier withholds supply, which on a single-source casting can stop the line. Receipt accruals are missed, so period material cost and creditor balances are misstated; purchase price variance never reaches the ledger, so standard cost quietly drifts away from reality and product costing becomes indefensible; and input tax credit is lost on mismatched documents.





6.1 Phased Implementation Roadmap

Phase	Focus	Scope delivered	Exit condition
Phase 1	Foundation	Master-data cleansing and Product Hub governance, including structure and work-definition validation; Financials; chart of accounts segments.	Clean, owned master data and a ledger that closes without spreadsheets.
Phase 2	Buy and store	Procurement with the four-stage approval flow; Inventory Management; Receiving with inspection.	Every purchase commitment is system-approved and every stock movement recorded.
Phase 3	Make and ship	Manufacturing work execution; Supply Planning; Order Management with Shipping; Cost Management with standard cost and variances.	Work in process and product cost are reported from the system, not estimated.
Phase 4	Optimise	Oracle HCM; Quality and Maintenance; OTBI and Oracle Analytics; then the future-scope items in Section 2.1.	Decisions taken from dashboards rather than from reconciled spreadsheets.

7. Conclusion

The design treats ERP as an operating-model decision rather than a software purchase. Five ranked modules cover where the factory creates and loses value; sixteen master-data objects establish the foundation that must exist before a single work order can be released; three high-frequency transactions show how routine activity flows across departments and modules; a four-stage mandatory approval flow puts control at the point of financial and source commitment; and two tightly integrated pairings hold the design together, since a broken interface is what turns an integrated system back into functional silos.

In digital-maturity terms this reaches only the level at which the factory has one trustworthy record of what it bought, consumed, built and shipped, and what each of those cost. That level is unglamorous, but it is the precondition for everything above it. The Industry 4.0 capabilities listed as future scope — machine-level OEE capture, scheduling optimisation, predictive maintenance, digital-twin simulation — all consume accurate item structures, work definitions and timely shop-floor transactions. A digital twin built on spreadsheet operation times would simulate a factory that does not exist. Getting the master data, the approval flow and the integration right first is therefore not a delay to digital transformation at Bburago Group; it is the first stage of it.